

6	(a)	Faraday's law of electromagnetic induction states that the magnitude of the induced e.m.f. E is directly proportional to the rate of change of the magnetic flux linkage.		1
	(b)	(i)	Rate of increase in area of loop ABDCA $= [3 - (-5)] \times 0.15 = 1.2 \text{ m}^2 \text{ s}^{-1}$ Induced e.m.f. $= \frac{d\phi}{dt} = (0.20)(1.2)$ $= 0.24 \text{ V}$ Alternatively, $\varepsilon = Blv_{AB} + Blv_{CD}$ $= 0.20(0.15)(3) + 0.20(0.15)(5)$ $= 0.24 \text{ V}$	1 1 1
		(ii)	[The magnetic flux $\phi = BA$.] Since the area of the loop is increasing, the magnetic flux is increasing. By Lenz's law, the induced current in the loop will create a magnetic field that is acting out of the paper to oppose the increasing flux. By the right hand grip rule, the induced current will flow in the anti-clockwise direction of ABDCA (Cannot use Fleming's Right Hand Rule as an explanation in this case as question specifies using Lenz's law.)	1 1
		(iii)	By Fleming's Left Hand rule, the induced current will lead to a magnetic force that is in opposite direction to the velocity of the rods. This will cause the rod to slow down.	1

5

			In order to keep the speed constant such that the net force is zero, an external force (equal and opposite) is required to counter the magnetic force.	1
--	--	--	--	---

7	(a)	1	The existence of a threshold (minimum) frequency below which no	1
---	-----	---	---	---